5.13.55

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Question: Given the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{pmatrix} \quad \text{and} \quad \mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{1}$$

and the relation $\mathbf{A}^{-1} = \frac{1}{6}(\mathbf{A}^2 + c\mathbf{A} + d\mathbf{I})$, then the value of c and d are

Solution :

The characteristic equation for $\mathbf{A}$is given by $\det(A - \lambda I) = 0$.

First, we construct the matrix $(\mathbf{A} - \lambda\mathbf{I})$:

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & -2 & 4-\lambda \end{pmatrix} \tag{2}$$

Next, we calculate its determinant:

$$\begin{aligned} \det(\mathbf{A} - \lambda\mathbf{I}) &= (1-\lambda) \begin{pmatrix} 1-\lambda & 1 \\ -2 & 4-\lambda \end{pmatrix} \\ &= (1-\lambda)\,[(1-\lambda)(4-\lambda) - (1)(-2)] \\ &= (1-\lambda)\left[4 - 4\lambda - \lambda + \lambda^2 + 2\right] \\ &= (1-\lambda)\left(\lambda^2 - 5\lambda + 6\right) \\ &= \lambda^2 - 5\lambda + 6 - \lambda^3 + 5\lambda^2 - 6\lambda \\ &= -\lambda^3 + 6\lambda^2 - 11\lambda + 6 \end{aligned}$$

Setting the determinant to zero gives the characteristic equation. Multiplying by -1 for convention, we get:

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0 \tag{3}$$

The theorem allows us to substitute the matrix A for λ in the characteristic equation. The constant term is multiplied by the identity matrix I.

$$\mathbf{A}^3 - 6\mathbf{A}^2 + 11\mathbf{A} - 6\mathbf{I} = \mathbf{O} \tag{4}$$

where $\mathbf{O}$ is the zero matrix.

We can rearrange Equation (4) to solve for the inverse. First, move the identity term to the right side:

$$\mathbf{A}^3 - 6\mathbf{A}^2 + 11\mathbf{A} = 6\mathbf{I} \tag{5}$$

Now, multiply the entire equation by $\mathbf{A}^{-1}$ from the left:

$$\mathbf{A}^{-1}\left(\mathbf{A}^3 - 6\mathbf{A}^2 + 11\mathbf{A}\right) = \mathbf{A}^{-1}(6\mathbf{I})$$

$$\mathbf{A}^2 - 6\mathbf{A} + 11\mathbf{I} = 6\mathbf{A}^{-1}$$

Finally, isolate $\mathbf{A}^{-1}$ by dividing by 6:

$$\mathbf{A}^{-1} = \frac{1}{6}\left(\mathbf{A}^2 - 6\mathbf{A} + 11\mathbf{I}\right) \tag{6}$$

We now compare the derived expression (6) with the expression given in the question.

Derived Expression: $\mathbf{A}^{-1} = \frac{1}{6}\left(\mathbf{A}^2 - 6\mathbf{A} + 11\mathbf{I}\right)$

Given Expression: $\mathbf{A}^{-1} = \frac{1}{6}(\mathbf{A}^2 + c\mathbf{A} + d\mathbf{I})$
By directly comparing the coefficients of the matrices $\mathbf{A}$ and $\mathbf{I}$ inside the parentheses, we find:

$$c = -6 \quad \text{and} \quad d = 11 \tag{7}$$

The solution is therefore third option, $(c, d) = (-6, 11)$.